

THOUFIQ YAHYA

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Senior Graphics & Embedded Software Engineer | Automotive HMI

EXECUTIVE SUMMARY

Senior Software Engineer and Technical Lead with 13+ years of experience developing graphics-intensive automotive infotainment systems, embedded software platforms, and Human Machine Interfaces (HMI). Expertise spans C++, Embedded Linux, Qt/QML, OpenGL ES, GLSL shader programming, Unity, Android-based platforms, and software architecture.

Proven track record of delivering production software for Mercedes-Benz MBUX and MB.OS platforms, leading cross-functional engineering teams, designing scalable architectures, and optimizing graphics performance on resource-constrained embedded hardware.

Strong background in real-time rendering, 3D asset pipelines, GPU programming, automotive UI frameworks, conversational AI systems, and Linux-based development environments. Combines deep technical expertise with leadership experience across graphics engineering, embedded systems, software architecture, and product delivery.

FUNCTIONAL EXPERTISE

Programming Languages

- C++
- Python
- JavaScript
- Qt/QML
- Bash

Graphics & Rendering

- OpenGL ES
- Shader Development
- Unity3D
- GLSL, HLSL
- Qt Rendering Framework
- Blender

Automotive

- Automotive IVI
- Voice User Interface (VUI)
- Human Machine Interface (HMI)

Embedded Systems

- Embedded Linux
- IPC (DBus)
- GCC Toolchain
- Docker
- CMake
- System Integration

Software Engineering

- Software Architecture
- SOLID Principles
- OOP
- TDD
- GMock
- UML
- Design Patterns
- CI/CD
- GTest

- Owned development and enhancement of an in-house automotive 3D rendering engine used to visualize vehicle assets within Mercedes-Benz infotainment systems.
- Developed and optimized rendering features using OpenGL ES, Qt Rendering Framework, GLSL, and HLSL shaders, targeting resource-constrained embedded automotive hardware.
- Designed and implemented advanced visual effects, rendering workflows, and GPU-accelerated shader-based features for next-generation HMI experiences.
- Built tooling and automation around Blender and GLTF asset pipelines, enabling efficient integration of high-quality 3D vehicle models into production infotainment environments.
- Collaborated with designers, graphics engineers, and platform teams to improve rendering performance, memory efficiency, and maintainability of the graphics stack.
- Evaluated and prototyped Unity-based rendering solutions for Android-powered automotive platforms, including architecture assessment, technology feasibility studies, and integration strategies.
- Investigated next-generation graphics technologies and rendering architectures to support future software-defined vehicle (SDV) initiatives.

Mercedes-Benz Research & Development India**April 2016 - September 2023 (7.5yrs)**

Led the India Widgets Team responsible for architecture, development, integration, and delivery of production MBUX widgets for Mercedes-Benz infotainment platforms.

- **Software Design & Architecture:** Collaborated with product owners, UX designers, and concept teams to transform requirements into scalable software architectures using UML and modern software engineering practices.
- **Technical Leadership:** Led a team of five engineers responsible for development, code reviews, technical mentoring, milestone planning, and successful feature delivery.
- **Qt/QML/C++ Development:** Directed implementation of complex UI widgets and reusable software components for the Mercedes-Benz MBUX ecosystem.
- **Architectural Excellence:** Applied SOLID principles, design patterns, and object-oriented design techniques to build maintainable and extensible software systems.
- **Cross-Team Collaboration:** Coordinated with platform, CI/CD, integration, and testing teams to streamline development workflows and deployment pipelines.
- **Quality & Delivery:** Ensured adherence to coding standards, software quality metrics, and project timelines across multiple releases.

L & Technology Services - Technical Consultant**June 2014 - April 2016 (2yrs)**

Developed embedded user interfaces for Cisco/NDS set-top box platforms.

- Built Electronic Program Guide (EPG) applications for embedded devices.
- Developed UI components using QML, JavaScript, and ActionScript.
- Optimized UI responsiveness and rendering performance on constrained hardware.
- Collaborated with hardware and platform teams on system integration and performance tuning.

I2Play Interactive - Game Developer**Nov 2012 – June 2014 (1.5 yrs)**

Worked on MMOG development with a focus on graphics systems and visual effects.

- Developed particle systems, animations, and visual effects.
- Contributed to an in-house game engine using OOP and design patterns.
- Implemented gameplay and rendering-related features.
- Led development of mini-games and interactive visual components.

Excel Goodies - Game Developer / Web Designer**June 2011 – Jan 2012 (6 months)**

Worked on educational games and interactive web applications.

- Developed 2D educational games and reusable UI components.
- Created graphics and visual assets using Photoshop and Inkscape.
- Developed web-based user interfaces using HTML and CSS.

AWARDS & RECOGNITION

- Star Performer — L&T Technology Services (2015)
 - Winner, Innovation Boot Camp — Mercedes-Benz R&D (2019)
 - Silver Star — Mercedes-Benz R&D (2020)
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EDUCATION

Bachelor of Information Technology**2006 - 2009**